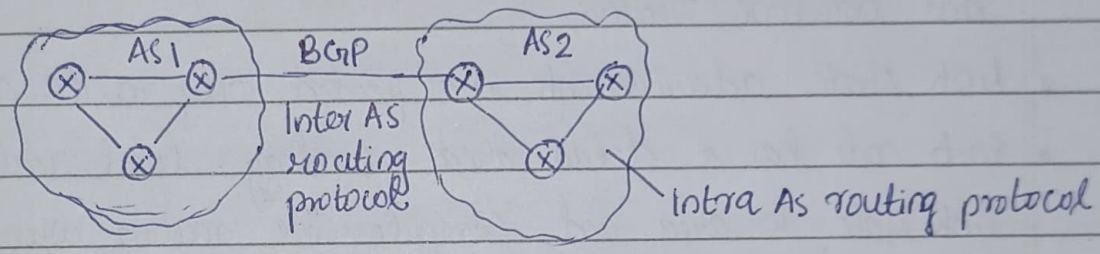
03/03/2026

Autonomous system

Group routers within a region

Autonomy -
Scale -



BGP - Border Gateway Protocol.

Intra AS routing Protocols

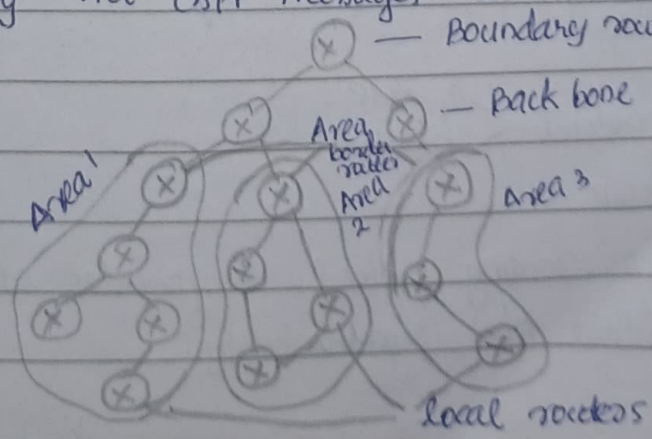
RIP : classic DV

EIGRP : Enhanced Interior Gateway

OSPF : link state.

OSPF

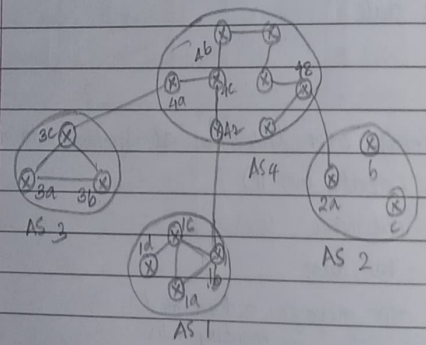
- Each Router floods OSPF link-state advertisements to all other routers in AS (directly over IP)
- Multiple link cost metric
  - bw, delay
- Security - All OSPF messages authenticated.



⇒ Autonomous system

→ Boundary routers → connected with other boundary routers for inter AS routing

* This is a 2 level hierarchical OSPF having local area routers and backbone routers.
* Link state advertisements are flooded only in the area or backbone
* Each node has a detail area topology. local routers flood the link state in area and compute the routing within the area.
* They will forward the packets outside via backbone routers.
* Every area → has area border routers
* Area border routers summarise the distance to the destinations in their own area and advertise it to the backbone.
* Backbone routers run OSPF and is limited only to the backbone.



Border Gateway Protocol < e-BGP, i-BGP

~~IGP~~ ~~I-BGP~~, ~~E-BGP~~

* The default inter AS routing protocol. It provides each AS a means to obtain n/w reachability.

- It allows us to determine the routes for AS to other n/ws -
- Based on both reachability, info and policy
- It also advertises the destination reachability info
- Propagate info to all internal routers (i-BGP)

BGP sessions:

* 2 BGP routers exchange messages over semi permanent ~~BGP~~ TCP connections.

* BGP messages are exchanges b/w peers over TCP connection and messages can be open, update (new path), KEEP ALIVE, NOTIFICATIONS (report errors)
(open tcp conn to remote BGP peer)

- RIP - max no. of nodes = 15
- EIGRP - used for cisco n/w.

Q. If direct broadcast address of the n/w is 168.17.07.255. what could be the network mask.

Ans: DBA - 168.17.07.255
↳ class B
n/w mask - 255.255.0.0 (for cls B)
Since broadcast add^n → 168.17.248.0

168.17.07.255 → host id
00000111
255
7
248

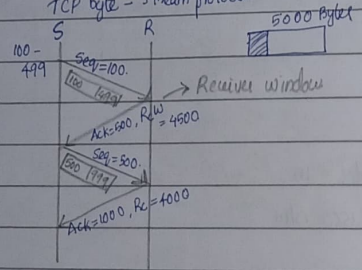
10/03/26

TCP Header:

32 bit

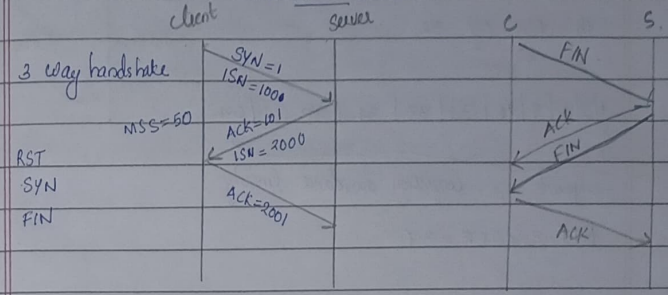
20-60 bytes	Source Port	Destination Port
	Sequence No.	
	Acknowledgement No.	
	Hdr length: 6 bit reserve; U R G, A C K, P S H, R S T, S Y N, F I N	window size (Advt window)
	checksum	urgent pointer
	options (0-40) bytes	
	Data	

TCP byte - stream protocol



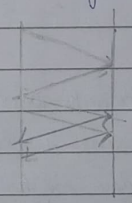
* Reliability of TCP
* flow control - By Receiver window
* Congestion control.
* Header length should be 20-60 bytes → 15 × 4 = 60

TCP Connection Establish



Source Port		Destination port
Sequence No.		
Acknowledgement No.		
Hdr length: 6 bit reserve; U R G, A C K, P S H, R S T, S Y N, F I N		window size
Checksum		urgent pointer
options		
data		

Congestion Control



Slow Start → starts by ~~sending~~ sending one segment. window size increases exponentialy until threshold.

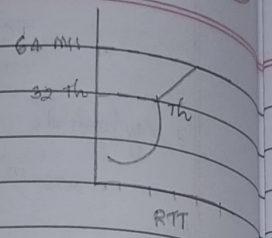
Assume $W_R = 64\,KB$

$1\,mss = 1\,KB$

$W_R = 64\,mss$

$W_R = 64 MSS$

$Th = \frac{W_R}{2} = \frac{64}{2} = 32.$



1 | 2 | 4 | 8 | 16 | 32 | 33 | 34 | 35 . . . | 64

↑ SS

exponent — congestion avoidance linearly

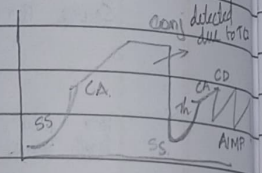
RTT = 32+5 = 37

2. Congestion Avoidance Phase

increase window linearly until max receiver window.

3. Congestion Detection Phase

SS — Severe congestion (- Pkt lost, - Ack lost) / mild congestion — CA



Q. Consider the effect of using SS on a line with 10ms and no congestion. The receiver window is 24 KB and the MSS is 2 KB. How long does it take before the first full window can be sent.

Ans. Receiver ~~size~~ size window max, $W_R = 24KB = \frac{24 \times 2}{2} = \cancel{48}\ 12$ MSS

MSS = 2 KB , RTT = 10ms

$Th = \frac{W_R}{2} = \cancel{\frac{48}{2}} = 24$

$Th = \frac{W_R}{2} = \frac{12}{2} = 6$

1 | 2 | 4 | 6 | 7 | 8 | 9 | 10 | 11 | 12

↑ SS ↑ Th 9 RTT

Q. Consider an instance of TCP's additive increase multiplicative decrease (AIMD) policy where the window size at the start of SS is 2 MSS and the threshold at the start of 1st transmission is 8 MSS. Assume Timeout occurs at 5th transmission. Find the congestion window at the end of 10th transmission.

Ans: Start at SS,

2 | 4 | 8 | 9 | 10 timeout
1 2 3 4 5

2 | 4 | 5 | 6 | 7 | 8
6 7 8 9 10

At the end of 10th transmission, congestion window is at 8 MSS.

~~Q. Consider an instance of TCP's additive increase multiplicative~~

11/03/26 Q. On a TCP connection current congestion window size is 4 KB. The window advertised by receiver is 6 KB. The last byte sent by sender is 10240 and the last byte acknowledged by receiver is 8192.

a) what is the current window size at sender.
b) what is the amt of free space sender will hv

Ans: $C_{win} = 4KB$, $R_{win} = 6KB$

a) $W_S = \min(C_{win}, R_{win})$

$= \min(4KB, 6KB) = 4KB$

b) no. of bytes $= \underset{\text{sent}}{10240} - \underset{\text{ack}}{8192}$

$= 2048$

$4KB = 4096$ $(1KB = 1024)$ → Total size

amt of free space $= 4096 - 2048 = 2048$

VLANs

- In VLANs, computers servers and networking devices are logically connected regardless of their physical location.
- Achieve improved security, traffic management and n/w is simple.
- All the broadcast n/w traffic are separated & isolated with a separate broadcast traffic for each of the routers.

Router → Routes from 1 protocol n/w to another n/w of same protocol.

TCP/IP — Router — TCP/IP

Gateway → Routes from 1 protocol n/w to another protocol n/w by converting

TCP/IP — Gateway — X.25

17/03/26

Module 1

Video Streaming and CDNs

- stream video traffic - major consumer of internet BW.
- Challenge - heterogeneity (diff hr diff capabilities)
- sol^n - distributed, appl^n level

- video - sequence of img displayed at const. rate.
  dig img - array of pixels.
- Coding - use redundancy within and b/w img to ↓ #bits used in encode img.
- * CBR - encoding rate fixed.
- * VBR - encoding rate changes as amt of spatial, temporal coding changes.

Streaming stored video

Video server → internet → Client.

Challenges.

- server to client BW will vary
- Packet loss, delay due to congestion will delay playout, poor quality
- playout time must match original timing.
- client interactivity: pause, rewind, jump through video.
- packets may be lost, retransmitted.

DASH (Dynamic Adaptive streaming over HTTP)

server

- divides file into multiple chunks → encoded at multiple diff rates
  → stored in diff files - replicated in CDN nodes.

Client

- periodically estimates server-to-client BW.
- consulting manifest, req one chunk at a time
  - → choose max coding rate sustainable given current BW.
  - → " diff " at diff pts in time & from diff servers.

Client distribution networks (CDNs)

Intelligence at client: client determines
- when to req chunk.
- what encoding rate
- where to req chunk

Streaming video - encoding + DASH + playout buffering

Client distribution networks (CDNs)

challenge: how to stream content to 100 of 1000 of simultaneous users

→ serve/store multiple copies of videos at multiple geographically distributed sites.

→ enter deep: push CDN servers deep into many access n/w.

→ bring home: smaller no. of larger clusters in POPs near access net.

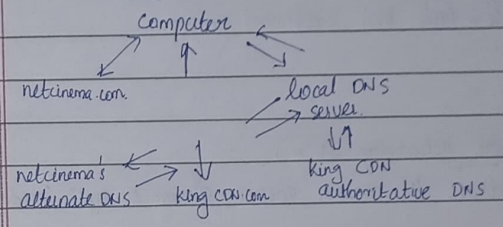
Netflix

- Stores copies of content at its OpenConnect CDN nodes.
- Subscriber req content, service provider returns manifest.

OTT challenges: ~~over~~ coping with a congested internet from the edge.

- what content to place
- from which CDN node to retrieve content

CDN content access



- get URL
- resolve URL via local DNS
- HTTP name for addrs
- to kingCDN author. DNS
- response from local DNS back to users computer

Module 3

~~MPLNs~~ MPLS

- A high-performance tech that directs data using labels instead of IP addresses.
- work b/w Layer 2 (data link) & layer 3 (n/w) - called Layer 2.5
- packets are assigned short labels; ~~not~~ routers fwd them based on ~~labels~~ labels - faster than traditional routing.

Data centre networking (DCN)

- n/wing infrastruce that connects servers, storage & appl^n inside a data centre.
- Features - high scalable, low latency, high available

Software Defined networking (SDN)

- A n/wing approach where the control plane is separated from the data plane.
- n/w is controlled by a centralized controller instead of individual devices.